

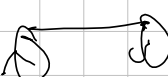
Abelian

$ab = ba \rightarrow \text{group } \mathbb{Z} \times \mathbb{Z}$



$S^1 \vee S^2$





$\frac{S^1 \vee S^1}{\mathbb{Z} * \mathbb{Z}}$



$T : \mathbb{Z} \times \mathbb{Z}$

$T \in \mathbb{Z} \times \mathbb{Z}$

$f : \mathbb{Z} \times \mathbb{Z}$

s_1

$s_1 \otimes s_2$

[illegible]

$\mathbb{R}/\mathbb{P}^2$: 0-cell: $\{v\}$
 1-cell: $\{x, y\}$
 2-cell: $\{z\}$

Hand-drawn sketches and text on a grid background:

- A circle with a wavy line inside, resembling a mouth or a stylized face.
- A flower-like shape with five petals.
- A heart shape with a small circle inside.
- A square with a small circle inside.
- Handwritten text: $\pi \times \pi \times \pi$, $\cup \pi \times \pi$, $\cup (0,0,2)$, $\pi \times \pi$, (π, π, π) .
- Arrows pointing from the text towards the sketches.

The sketches include:

- A series of five small circles arranged in a horizontal row.
- A flower-like shape with five petals.
- A spiral shape starting from a point and winding outwards.
- A horizontal line with four tick marks.
- A coordinate system with a curve starting from the origin and moving upwards and to the right.
- A region bounded by a curve, with arrows pointing outwards from the boundary.